Auteur: Michael Livchits
Studierichting: Informatica
Oefeningenreeks 5B nr. 1.11

$$I = \int_{-\infty}^{1} \frac{dx}{x^3}$$

Eerst vinden we de onbepaalde integraal

$$\int \frac{dx}{x^3} = -\frac{1}{2x^2} + C$$

Neem de limiet

$$I = \lim_{b \to 0-} \left[-\frac{1}{2x^2} \right]_{-\infty}^{b} + \lim_{b \to 0+} \left[-\frac{1}{2x^2} \right]_{b}^{1}$$

Eerst linker limiet:

$$I_1 = \lim_{b \to 0-} \left[-\frac{1}{2x^2} \right]_{-\infty}^{b} = \lim_{b \to 0-} \left(-\frac{1}{2b^2} + 0 \right) = -\infty$$

Nu rechter limiet:

$$I_2 = \lim_{b \to 0+} \left[-\frac{1}{2x^2} \right]_{b}^{1} = \lim_{b \to 0+} \left(-\frac{1}{2} + \frac{1}{2b^2} \right) = +\infty$$

$\Rightarrow$ divergeert

References